Paper questions E.5: Fusion and stars [67 marks]

1. [Maximum mark: 1]

What is the sequence for the evolution of a main sequence star of about 2 solar masses?

A. Red super giant → supernova → neutron star

B. Red giant → planetary nebula → white dwarf

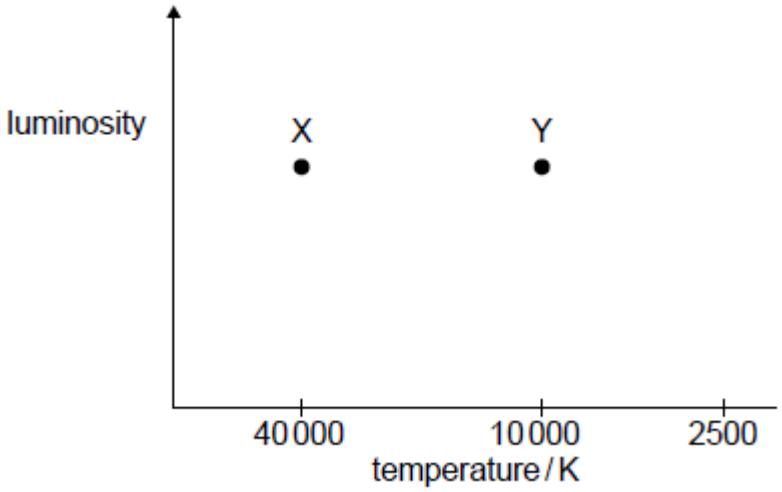
C. Red giant → supernova → white dwarf

D. Red super giant → planetary nebula → neutron star [1]

Markscheme
B

2. [Maximum mark: 1]

The Hertzsprung–Russell diagram shows two stars, X and Y.



What is $\frac{\text{radius of X}}{\text{radius of Y}}$?

A. $\frac{1}{16}$

B. $\frac{1}{4}$

C. 4

D. 16

[1]

Markscheme
A

3. [Maximum mark: 1]

Which process is the primary energy source in a red giant star?

A. gravitational contraction

B. nuclear fusion of hydrogen in the core

C. nuclear fusion of helium in the core

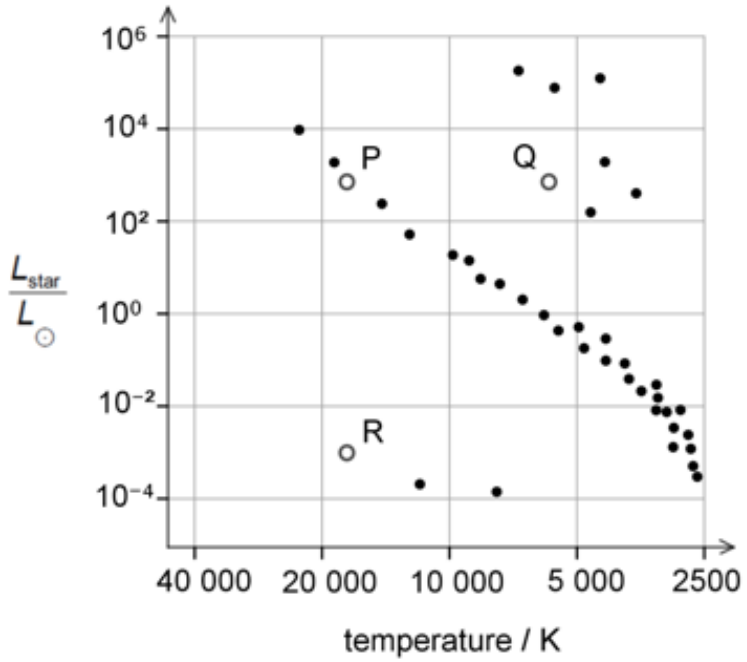
D. nuclear fusion of helium in the shell surrounding the core

[1]

Markscheme
C

4. [Maximum mark: 1]

Three stars P, Q and R are plotted in the Hertzsprung-Russel diagram with empty circles.



Which of the following lists the stars in the order of increasing radius?

A. Q, R, P

B. P, R, Q

C. R, Q, P

D. R, P, Q

[1]

Markscheme
D

5. [Maximum mark: 1]

Star X has the same surface temperature as the Sun and a luminosity of $10^4 L_\odot$

What is $\frac{\text{radius of X}}{\text{radius of the Sun}}$?

A. 10

B. 10^2

C. 10^3

D. 10^4

[1]

Markscheme
B

6. [Maximum mark: 1]

A star is on the main sequence.

What are the most abundant element(s) in the core of the star and in the outer layer of the star?

	Most abundant element(s) in the core	**Most abundant element(s) in the outer layer**
A.	Helium and lithium	Hydrogen
B.	Hydrogen	Helium
C.	Hydrogen and helium	Hydrogen and lithium
D.	Hydrogen	Helium and beryllium

[1]

Markscheme
B

7. [Maximum mark: 1]

A star has a radius 13 times that of the Sun and a luminosity that is 400 000 times that of the Sun.

The surface temperature of the Sun is 5700 K.

What is the surface temperature of the star?

A. 4.0×10^4 K

B. 7.6×10^4 K

C. 1.0×10^5 K

D. 1.4×10^7 K [1]

Markscheme
A

8. [Maximum mark: 1]

What is the likely evolutionary outcome for a star with the same mass as the Sun?

A. main sequence star → red giant → supernova → white dwarf

B. main sequence star → red giant → planetary nebula → white dwarf

C. main sequence star → super giant → supernova → neutron star

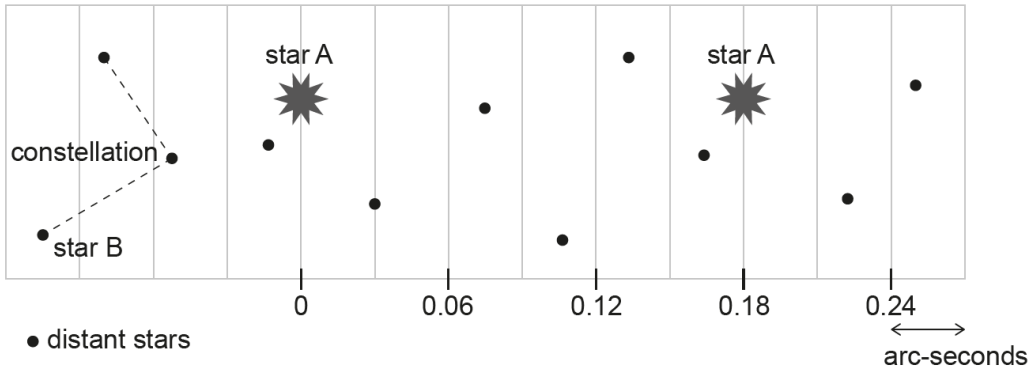
D. main sequence star → super giant → planetary nebula → black hole [1]

Markscheme

B

9. [Maximum mark: 12]

The diagram shows the extreme positions of star A six months apart as seen from Earth. The black dots are the fixed positions of distant stars. The horizontal scale is in arc-seconds.



(a.i) Show that the distance to star A is about 10 pc. [2]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

parallax angle = $\frac{0.18}{2}$ ✔

distance =« $\frac{1}{0.09}$ =» 11 «pc» ✔

Allow ECF for MP2.

For MP2, working or answer to at least 2 significant figures must be seen.

(a.ii) Calculate the ratio $\frac{\text{distance to star A}}{\text{distance from Earth to Sun}}$. [2]

Markscheme

$11 \times 3.26 \times 9.46 \times 10^{15} = 3.4 \times 10^{17}$ «m» ✔

$\frac{3.4 \times 10^{17}}{1.50 \times 10^{11}}$ ***OR*** 2.3×10^{6} ✔

Award [2] for BCA from interval 2.0×10^{6} to 2.4×10^{6}

(b) Star B is part of a constellation. Outline what is meant by a constellation. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

group of stars that appear to be close to one another/form a shape/pattern ✔

but are not gravitationally bound
OR
are not necessarily close in space ✔

(c) The analysis of the stellar spectrum of star B shows that it has a surface temperature of 9900 K. Star B is a main sequence star with a luminosity of 25 $L_{\odot}$, where $L_{\odot}$ is the luminosity of the Sun.

(c.i) Outline how the temperature of a star can be determined from its stellar spectrum.

[2]

Markscheme

identify peak wavelength ✔

use peak wavelength « in Wien's law $\lambda_{max}T = 2.9 \times 10^{-3}$ » to get T ✔

(c.ii) Predict the likely **final stage** in the stellar evolution of star B. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

use of mass-luminosity relationship to get $M = 2.5M_{SUN}$ ✔

white dwarf ✔

Do not allow use of temperature or wavelength, without considering the mass.

Do not allow ECF from MP1.

(d) Another star in the constellation is a Cepheid variable. Outline the reason for the variation in brightness of this star. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

surface area/radius/size/temperature of the star increases and decreases
OR
the star expands and contracts periodically ✔

as it expands, it cools/brightness decreases
OR
as it contracts, the temperature raises/brightness increases ✔

10. [Maximum mark: 11]
Alpha Centauri A and B form the closest binary star system to the Earth. This binary star system is the third brightest stellar object in the night sky.

(a) Arcturus is the fourth brightest star in the night sky. Its apparent brightness is slightly greater than that of the star Alpha Centauri A.

Outline why the binary star system Alpha Centauri appears brighter than Arcturus. [1]

Markscheme

** This question is from an examination for a previous syllabus. Please note that it may contain minor differences in marking or structure or contain parts that are not a match to the new syllabus (first assessment 2025).*

addition of Alpha Centauri B to the luminosity/brightness of the binary star system
OR
combined luminosity/brightness of the binary stars ✔

OWTTE

(b) Alpha Centauri A is a main sequence star.

The following data are available:

Parallax angle of Alpha Centauri A = 0.76 arcsec
Luminosity of Alpha Centauri A = 5.7×10^{26} W
Luminosity of the Sun $L_{\odot} = 3.8 \times 10^{26}$ W

(b.i) Show that the distance from Alpha Centauri A to our solar system is about 4×10^{16} m. [2]

Markscheme

$d = \frac{1}{0.76}$ ***OR*** 1.3 «pc» ✔

$d = \frac{3.26 \times 9.46 \times 10^{15}}{0.76}$ ***OR*** d = 4.1×10^{16} «m» ✔

MP2 scores MP1.

Answer given, so award MP2 if candidates show a correct full substitution ***OR*** *the value with an extra significant figure*

(b.ii) Calculate the apparent brightness of Alpha Centauri A. [1]

Markscheme

«$b = \frac{5.7 \times 10^{26}}{4\pi(4.1 \times 10^{16})^2}$ =» 2.7×10^{-8} «W/m» ✔

Accept use of 4×10^{16}

(b.iii) Show that the mass of Alpha Centauri A is 1.1 $M_{\odot}$, where $M_{\odot}$ is the mass of the Sun. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

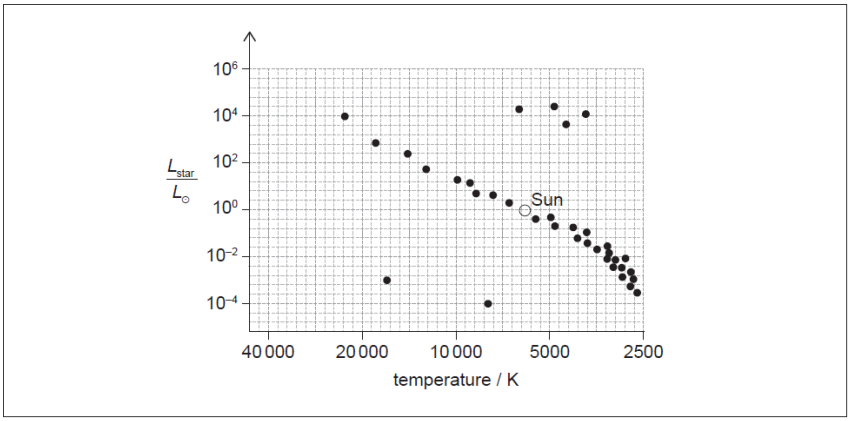
Use of $L = M^{3.5}$ ✔

$M = \left(\frac{5.7}{3.8}\right)^{\frac{1}{3.5}} M_{\odot}$ ***OR*** $M=1.12M_{\odot}$ ✔

*Answer given, so award MP2 if candidates show a correct full substitution **OR** the value with an extra significant figure*

(c) On the Hertzsprung-Russell (HR) diagram,

(c.i) plot, with an X, the position of Alpha Centauri A;



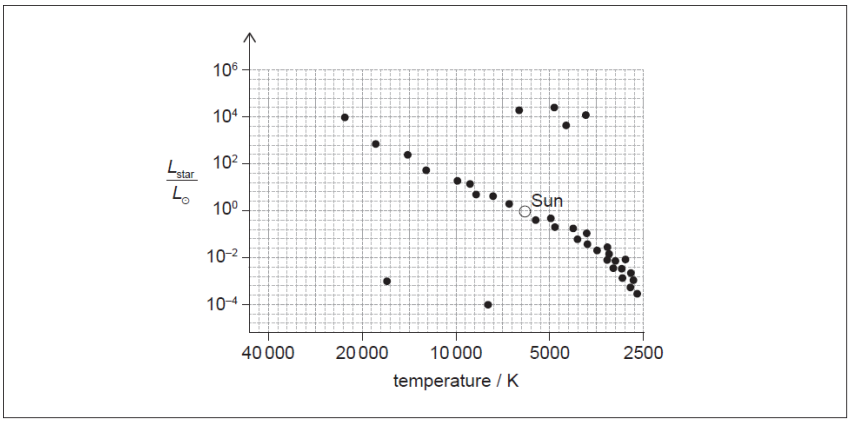
[1]

Markscheme

position of A correct (just above and to the left of the Sun) ✔

Allow a position from the Sun up to the second drawn star to its left.

(c.ii) draw the most likely future evolutionary path of Alpha Centauri A, labelling the two main regions of this path.



[2]

Markscheme

loop via red giants to white dwarves

OR

line towards red giants, then white dwarves ✔

red giants and white dwarves' approximate regions labelled ✔

Allow areas on the upper right-hand and lower left-hand region.

(d) Explain how a cepheid variable star can be used for the measurement of very large astronomical distances. [2]

Markscheme

*** This question part has been identified as not part of the new syllabus (first assessment 2025).*

Cepheid variable stars have luminosity that varies «with time»

OR

luminosity of a Cepheid is determined from its «brightness» period ✔

apparent brightness is measured

OR

use of $b = L/4\pi d^2$ ✔

«allows the calculation of distance»

Accept T = f(L)

OWTTE

11. [Maximum mark: 13]

The parallax angle of the star Vega is 0.131 arc seconds.

(a.i) Describe what is meant by parallax angle. [1]

Markscheme

the angle at which the star subtends 1 AU

OR

half the angle between star and Earth measured six months apart ✔

Accept a sketch that identifies the correct angle, i.e., half the angle subtended from star to Earth at both sides of Sun.

(a.ii) Show that the distance to Vega is 25 light years. [1]

Markscheme

$d = \frac{1}{0.131} x 3.26$ ***OR*** $d = 24.9$ «ly» ✔

(b) The following information is available for the stars Vega and β Ori.

	Distance / ly	Luminosity	Temperature / K
Vega	25	$54\,L_\odot$	9600
β Ori	780	$40\,000\,L_\odot$	11 000

$L_\odot$ is the luminosity of our Sun.

(b.i) Determine whether Vega or β Ori appears brighter from Earth. [3]

Markscheme

$$\frac{b_v}{b_\beta} = \ll \frac{L_v}{L_\beta}\left(\frac{d_\beta}{d_\beta}\right)^2 = \gg \frac{54}{40000}\left(\frac{780}{25}\right)^2 =$$

OR

calculates one brightness in terms of $L_\odot$ ✔

$\frac{b_v}{b_\beta} = 1.3$

OR

calculates a second brightness in terms of $L_\odot$ ✔

so Vega appears brighter ✔

*Award **[0]** for a bald correct answer.*

(b.ii) Calculate the ratio $\frac{\text{radius of β Ori}}{\text{radius of Vega}}$. [3]

Markscheme

$$\frac{L_\beta}{L_V} = \frac{40000}{54} = \ll \left(\frac{R_\beta}{R_V}\right)^2\left(\frac{11000}{9600}\right)^4 = \gg$$ ✔

$$\frac{R_\beta}{R_V} = \sqrt{\frac{40000}{54}\left(\frac{9600}{11000}\right)^4}$$ ✔

= 21 ✔

(c) Vega is a main-sequence star with a mass of about three solar masses.

(c.i) Vega is a main sequence star. Show that the mass of Vega is about three solar masses. [1]

Markscheme

$\frac{M_V}{M_\odot} = 54^{\frac{1}{3.5}}$ OR $\frac{M_V}{M_\odot} = 3.1$ ✔

(c.i) Discuss whether β Ori is a main sequence star. [2]

Markscheme

luminosity of β Ori is too high **AND**
temperature is comparable to Vega / mass «if main sequence would be» 20.6 $M_\odot$ ✔
so it cannot be a main sequence star ✔

MP2 *requires* ***MP1***.
If AND not present, i.e., if only one of the three features is mentioned, with conclusion present, award [1] max.

(c.ii) Describe the most likely final stage in the evolution of Vega. [2]

Markscheme

Vega is a low mass star / remnant will be lower than Chandrasekhar limit / it will go through red giant phase ✔
will evolve to become a white dwarf ✔
a very small/dense/hot/dead/low luminosity star ✔

12. [Maximum mark: 13]

(a) Identify with ticks [✓] in the table, the forces that can act on electrons and the forces that can act on quarks.

	Weak nuclear	Strong nuclear
Electrons		
Quarks		

[2]

Markscheme

Weak nuclear: 2 ticks ✓
Strong nuclear: quarks only ✓

(b) The following data is available for atomic masses for the fusion reaction

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + {}^{1}_{0}\mathrm{n}:$$

$^{2}_{1}\mathrm{H}$	2.0141 u
$^{3}_{1}\mathrm{H}$	3.0160 u
$^{4}_{2}\mathrm{He}$	4.0026 u

(b.i) Show that the energy released is about 18 MeV. [2]

Markscheme

«⊠» = 2.0141 + 3.0160 – (4.0026 + 1.008665) «= 0.0188 u»

OR

In MeV: 1876.13415 + 2809.404 – (3728.4219 + 939.5714475) ✓

= 0.0188 × 931.5 ***OR*** = 17.512 «MeV» ✓

*Must see either clear substitutions or answer to at least 3 s.f. for **MP2**.*

(b.ii) Estimate the specific energy of hydrogen by finding the energy produced when 0.4 kg of $^{2}_{1}\mathrm{H}$ and 0.6 kg of $^{3}_{1}\mathrm{H}$ undergo fusion. [2]

Markscheme

ALTERNATIVE 1

0.40 kg of deuterium is «$\frac{400}{2} \times 6.02 \times 10^{23}$» = 1.2×10^{26} nuclei
« 0.60 kg of tritium is the same number » ✓

So specific energy «$\frac{1.2\times10^{26}\times17.51\times10^{6}\times1.6\times10^{-19}}{0.4+0.6}$» = 3.4×10^{14} «J kg^{-1}»

ALTERNATIVE 2

«$17.51 \times 10^{6} \times 1.6 \times 10^{-19}$ =» 2.8×10^{-12} «J»

AND

«$(2.0141 + 3.0160) \times 1.66 \times 10^{-27}$ =» 8.35×10^{-27} ✓

«$\frac{2.8\times10^{-12}}{8.35\times10^{-37}}$» = 3.4×10^{14} «J kg^{-1}»

Allow ~2.1×10^{27} MeV kg^{-1} *for **MP2**.*

*Allow **ECF** from **MP1** for both **ALTs**.*

(c) It is hoped that nuclear fusion can be used for commercial production of energy.

Outline

(c.i) **two** difficulties of energy production by nuclear fusion. [2]

Markscheme

Requires high temp/pressure ✓
Must overcome Coulomb/intermolecular repulsion ✓
Difficult to contain / control «at high temp/pressure» ✓
Difficult to produce excess energy/often energy input greater than output / ***OWTTE*** ✓
Difficult to capture energy from fusion reactions ✓
Difficult to maintain/sustain a constant reaction rate ✓

(c.ii) **one** advantage of energy production by nuclear fusion compared to nuclear fission. [1]

Markscheme

Plentiful fuel supplies ***OR*** larger specific energy ***OR*** larger energy density ***OR*** little or no «major radioactive» waste products ✓

Allow descriptions such as "more energy per unit mass" or "more energy per unit volume"

(d) Tritium ($^{3}_{1}\mathrm{H}$) is unstable and decays into an isotope of helium (He) by beta minus decay with a half-life of 12.3 years.

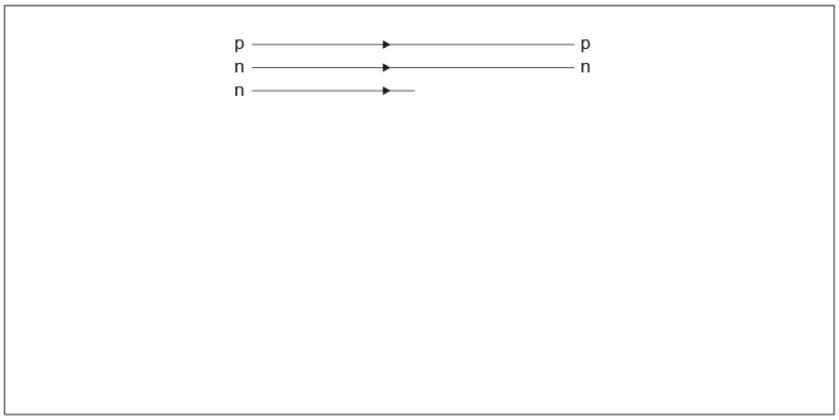
(d.i) State the nucleon number of the He isotope that $^{3}_{1}\mathrm{H}$ decays into. [1]

Markscheme

3 ✓

*Do **not** accept* $^{3}_{2}He$ *by itself.*

(d.ii) The following diagram is an incomplete Feynman diagram describing the beta minus decay of $^{3}_{1}\mathrm{H}$ into He. Complete the diagram and label all the missing particles.



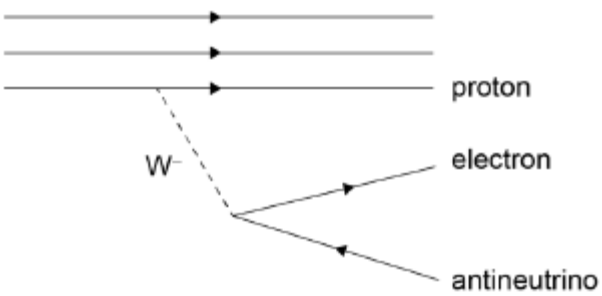
[3]

Markscheme

Proton shown ✓

W- shown ✓

Produces electron/e^- /$\boxtimes^-$ and antineutrino / $\overline{\upsilon}$ with proper arrow directions. ✓



Allow solid, dashed, or wavy line for W-particle.

Must see bar on antineutrino if symbol used.

13. [Maximum mark: 1]

State the main element that is undergoing nuclear fusion in star C.

[1]

Markscheme

Hydrogen ✓

14. [Maximum mark: 2]

Determine which star will appear to move more.

[2]

Markscheme

Star Y ✓

because parallax angle is greater ***OR*** star Y is closer «and that means movement relative to distant stars is greater» ✓

Allow reverse argument for star X

15. [Maximum mark: 1]

State the main element that is undergoing nuclear fusion in star C.

[1]

Markscheme

Hydrogen ✓

16. [Maximum mark: 1]

A fusion reaction of one nucleus of hydrogen-2 and one nucleus of hydrogen-3 converts 0.019 u to energy. A fission reaction of one nucleus of uranium-235 converts a mass of 0.190 u to energy.

What is the ratio $\frac{\text{specific energy of this fusion of hydrogen}}{\text{specific energy of this fission of uranium}}$?

A. 0.1

B. 0.2

C. 5

D. 10 [1]

Markscheme

C

17. [Maximum mark: 1]

Carbon (C-12) and hydrogen (H-1) undergo nuclear fusion to form nitrogen.

$^{12}_{6}\mathrm{C} + ^{1}_{1}\mathrm{H} \rightarrow \mathrm{N} +$ photon

What is the number of neutrons and number of nucleons in the nitrogen nuclide?

	Number of neutrons	Number of nucleons
A.	7	13
B.	6	13
C.	6	7
D.	7	6

[1]

Markscheme

B

18. [Maximum mark: 2]

The astronomical unit (AU) and light year (ly) are convenient measures of distance in astrophysics. Define each unit.

AU:

ly:

[2]

Markscheme

AU: «average» distance from the Earth to the Sun ✓

ly: distance light travels in one year ✓

19. [Maximum mark: 1]

Show that the apparent brightness $b \propto \frac{AT^4}{d^2}$, where d is the distance of the object from Earth, T is the surface temperature of the object and A is the surface area of the object.

[1]

Markscheme

substitution of $L = \sigma AT^4$ into $b = \frac{L}{4\pi d^2}$ giving $b = \frac{\sigma AT^4}{4\pi d^2}$

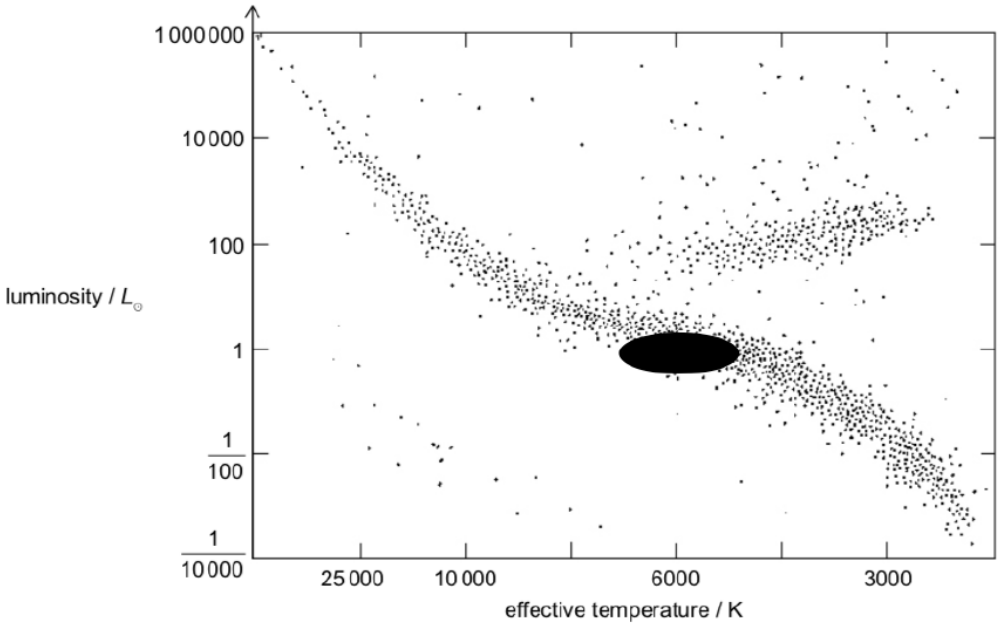
Removal of constants σ and 4π is optional

20. [Maximum mark: 1]

Identify, on the HR diagram, the position of the Sun. Label the position S. [1]

Markscheme

the letter S should be in the region of the shaded area ✔



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